E-Commerce Sales & Customer Behavior Analysis

COMPREHENSIVE EXPLORATORY DATA ANALYSIS (EDA) REPORT

1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) conducted on the UCI Online Retail Dataset, representing transactional data for a UK-based non-store online retail company. The primary objective of this investigation is to decode customer purchasing behaviors, isolate the most profitable product lines, understand geographic revenue distributions, and identify temporal demand shifts. These insights are designed to support data-driven inventory management, marketing resource allocation, and customer retention strategies.

The analysis successfully processed over 500,000 transaction records. The findings reveal a business heavily reliant on Q4 seasonality, a clear divide between standard retail and high-volume wholesale customer cohorts, and a healthy transaction completion rate with minimal cancellations. By leveraging these patterns, the organization can optimize its supply chain for peak periods and formally segment its marketing efforts to maximize Customer Lifetime Value (CLV).

TOTAL PROCESSED RECORDS	TRANSACTION COMPLETION RATE	PEAK SALES MONTH
541,909	**97.8%**	**November**

2. Data Processing & Methodology

To ensure the integrity of the analysis, a rigorous data cleaning and preprocessing pipeline was executed prior to the exploratory phase. The raw dataset contained over half a million rows, which required significant formatting to extract accurate business metrics.

- **Missing Value Treatment:** The dataset contained null values primarily within the `CustomerID` column. Because this analysis heavily weighs customer behavior, segmentation, and frequency, records lacking a `CustomerID` were isolated and removed to prevent skewed cohort analytics.
- **Feature Engineering:** Several new dimensions were derived to enable deep-dive analytics. A `TotalPrice` metric was calculated by multiplying `Quantity` by `UnitPrice`. Furthermore, the `InvoiceDate` was parsed to extract `YearMonth`, `DayOfWeek`, and `Hour`, facilitating temporal trend analysis.

- **Cancellation Handling:** Cancelled orders (denoted by an 'C' prefix in the `InvoiceNo`) and transactions with negative quantities were separated from the main sales dataset. This separation ensures that revenue calculations and top-selling product metrics reflect actual completed sales rather than gross attempts.

3. In-Depth Exploratory Findings



Figure 1: Compilation of E-Commerce EDA plots and distributions.

The analysis was guided by ten core business questions. The findings have been grouped into logical categories detailing product performance, geographic distribution, temporal trends, customer behavior, and transaction anomalies.

3.1 Product Performance and Demand (Questions 1, 4, 9)

Understanding which products drive the business is critical for inventory and supply chain management. The analysis evaluated products by both total revenue generated and sheer volume moved.

Highest Revenue Generators: The product "PAPER CRAFT, LITTLE BIRDIE" stands out as the single most lucrative item in the inventory, followed closely by the "REGENCY CAKESTAND 3 TIER" and the "WHITE HANGING HEART T-LIGHT HOLDER". These items consistently demonstrate high demand and command a price point that maximizes overall revenue yield.

Highest Quantity Sold: When evaluating pure volume, "PAPER CRAFT, LITTLE BIRDIE" again dominates, indicating it is not just a high-margin item but a massive bulk-seller. The "MEDIUM CERAMIC TOP STORAGE JAR" also moves in massive quantities, though it yields slightly lower total revenue compared to the cakestands.

Demand Shifts: Demand is highly seasonal and heavily skewed toward the latter half of the year. Products categorized under gifting, crafts, and home decorations experience a massive surge in demand leading up to Q4. The data indicates that inventory turnover for these specific categories accelerates rapidly starting in September.

Rank	Top Revenue Products	Top Volume (Quantity) Products
1	PAPER CRAFT, LITTLE BIRDIE	PAPER CRAFT, LITTLE BIRDIE
2	REGENCY CAKESTAND 3 TIER	MEDIUM CERAMIC TOP STORAGE JAR

Rank	Top Revenue Products	Top Volume (Quantity) Products
3	WHITE HANGING HEART T-LIGHT HOLDER	WORLD WAR 2 GLIDERS ASSTD DESIGNS
4	JUMBO BAG RED RETROSPOT	JUMBO BAG RED RETROSPOT
5	MEDIUM CERAMIC TOP STORAGE JAR	WHITE HANGING HEART T-LIGHT HOLDER

3.2 Geographic Revenue Distribution (Question 2)

The dataset reveals that the United Kingdom completely dominates total sales, which aligns with the company's domestic operational base. However, to understand expansion opportunities, the UK data was isolated to evaluate the true performance of international markets.

Excluding the UK, the top international revenue contributors are the **Netherlands, EIRE (Ireland), Germany, and France**. The Netherlands represents a uniquely dense market; despite having fewer overall transactions than Germany or France, the average order value from the Netherlands is extraordinarily high, suggesting a B2B or wholesale distribution network operating within that specific region.

3.3 Temporal Sales Trends (Question 3)

Time-series analysis of the parsed `InvoiceDate` reveals highly actionable patterns regarding when customers are most likely to convert.

Monthly Trends: Sales grow steadily but slowly through the first three quarters of the year. In September, a steep upward trajectory begins, peaking dramatically in **November**. This indicates a business model heavily reliant on holiday shopping and Q4 consumer spending. Surprisingly, December shows a sharp drop-off, likely due to shipping cut-off dates for holiday delivery.

Weekly and Hourly Trends: On a micro level, the purchasing week is heavily front-loaded. **Thursday** experiences the highest aggregate order volume of any day. When mapping order volume by the hour, a perfect bell curve emerges centered exactly on midday. Order activity begins at 8:00 AM, peaks sharply at **12:00 PM (Noon)**, and tapers off almost entirely by 6:00 PM. This suggests a customer base that shops primarily during working hours or on their lunch breaks.

3.4 Customer Behavior and Order Value (Questions 5, 6)

Average Order Value: The distribution of the total price per line item is heavily right-skewed. The vast majority of individual item purchases fall strictly under $20.00. The density of transactions rapidly decays as the price increases, confirming that the company is primarily a high-volume, low-margin retailer.

Valuable Customer Segments: A scatter plot analysis of Customer Order Frequency versus Total Spend reveals two distinct customer cohorts. The primary cohort is the standard retail shopper, characterized by 1 to 5 invoices and a total lifetime spend under $2,000. However, the data exposes a secondary, highly valuable cohort of B2B/wholesale customers. These individuals purchase very frequently (generating 15 to 40+ unique invoices) and routinely spend between $6,000 and $10,000+. This small segment of users drives a disproportionate amount of total revenue.

3.5 Anomalies, Elasticity, and Cancellations (Questions 7, 8, 10)

Cancellations: Assessing the health of the fulfillment pipeline, the data shows that only **2.2%** of all recorded transactions are cancellations. This is an exceptionally healthy completion rate (97.8%), suggesting that stock-outs, payment failures, and buyer's remorse are not critical threats to current revenue streams.

Outliers & Unusual Transactions: The dataset contains massive outliers in both quantity and unit price. The most notable anomalies are single wholesale orders containing tens of thousands of units (e.g., a single order of over 80,000 "PAPER CRAFT" units). While technically outliers in a statistical sense, these are valid, business-critical transactions that represent massive wholesale liquidations.

Price vs. Quantity Elasticity: There is a distinct, classic negative price elasticity curve observed in the dataset. A bivariate scatter plot analyzing unit price against quantity purchased clearly demonstrates that bulk quantities (orders of 100+ units per line item) are exclusively restricted to lower-cost items priced under $5.00. As the unit price increases beyond $10.00, the maximum quantity purchased per order drops to single digits.

4. Strategic Business Recommendations

By synthesizing the exploratory data analysis, several key operational and strategic shifts are recommended to maximize profitability and streamline operations:

- **Aggressive Q3 Inventory Stockpiling:** Given the massive revenue spike concentrated entirely in November, the supply chain must be fully stocked with top items (e.g., Paper Craft, Regency Cakestands, Storage Jars) no later than mid-October. The sharp drop in December suggests the company cannot afford backorders during the November rush, as late deliveries will miss the holiday window entirely.
- **Temporal Marketing Optimization:** Marketing budgets and email campaigns are currently deployed uniformly. Based on the hourly and daily heatmaps, digital ad spend and promotional email blasts should be heavily re-weighted. Campaigns should be scheduled to hit inboxes on Thursday mornings between 10:00 AM and 11:30 AM to capture the proven midday (Noon) purchasing surge.
- **Formalize a B2B Wholesale VIP Program:** The data proves the existence of a highly lucrative, high-frequency wholesale cohort that drives significant revenue through bulk, low-unit-price orders (as seen in the Netherlands and the UK). The company should formally segment these users into a B2B VIP tier, offering tiered bulk-discount pricing and dedicated account managers to retain these crucial accounts and encourage even larger bulk acquisitions.
- **Expand European Hub Operations:** The disproportionately high order values originating from the Netherlands and EIRE indicate strong product-market fit outside the UK. Leadership should investigate establishing localized fulfillment hubs or dedicated shipping lanes to these regions to reduce transit times and capitalize on the existing wholesale demand.